

FREDERICK, OK, USA

WMO: 723528

Lat: 34.344N

Lon: 98.983W

Elev: 383

StdP: 96.81

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 03981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-5.7	-16.0	1.0	-3.9	-12.9	1.3	-0.4	14.5	7.8	12.8	7.9	4.6	10	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	40.2	22.6	38.7	22.8	37.4	22.8	25.6	34.4	25.0	33.4	24.4	32.6	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	19.0	28.6	22.7	18.3	27.8	22.3	17.8	27.6	81.1	34.8	78.0	33.4	75.6	32.5	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.2	10.1	DB	-11.6	41.7	2.3	1.9	-13.2	43.1	-14.5	44.2	-15.8	45.2	-17.5	46.6	(4)
(5)			WB	-12.4	26.9	2.2	2.1	-14.0	28.4	-15.3	29.6	-16.5	30.7	-18.0	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	5.7	7.5	12.6	17.3	22.4	27.5	29.5	29.3	24.9	18.2	11.7	6.0
	DBStd	9.73	5.23	5.87	5.44	4.59	4.39	3.02	2.67	2.81	3.69	4.83	5.11	5.05
	HDD10.0	484	152	107	35	5	0	0	0	0	0	4	39	142
	HDD18.3	1624	393	306	189	71	15	0	0	0	2	62	203	383
	CDD10.0	3321	17	36	116	223	385	526	604	597	447	259	92	18
	CDD18.3	1419	0	1	12	39	142	276	346	339	199	59	6	0
	CDH23.3	17736	9	31	155	478	1551	3437	4741	4466	2171	619	74	5
	CDH26.7	9743	0	6	38	164	732	1927	2857	2674	1103	228	14	1
Wind	WSAvg	5.0	4.8	5.2	5.5	5.8	5.4	5.5	4.5	4.3	4.4	4.9	4.9	4.7
Precipitation	PrecAvg	713	23	31	48	53	110	90	51	69	92	73	38	28
	PrecMax	1139	111	103	143	137	296	270	124	201	214	286	132	132
	PrecMin	467	0	2	2	1	7	12	0	0	0	3	0	0
	PrecStd	162	25	25	35	37	73	62	35	54	61	71	34	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	27.3	30.7	34.0	38.1	41.2	42.0	42.3	39.2	33.8	28.7	23.3
		MCWB	12.2	14.9	16.4	17.6	20.9	22.5	22.1	22.4	21.6	20.4	17.6	12.6
	2%	DB	21.0	23.3	27.2	31.1	35.8	38.5	39.9	40.1	36.9	31.5	25.5	19.8
		MCWB	11.1	13.5	16.6	18.4	21.4	22.8	22.9	22.7	22.4	19.9	16.0	12.0
	5%	DB	17.6	20.3	24.8	28.0	33.0	37.0	38.4	38.5	34.6	29.1	22.8	17.1
		MCWB	9.4	12.2	15.8	17.8	21.0	22.9	23.1	22.9	22.1	19.1	14.8	10.6
	10%	DB	14.5	17.4	22.2	26.1	31.1	35.0	37.2	37.2	32.6	27.2	20.2	14.1
		MCWB	8.3	10.7	14.5	17.2	20.7	23.1	23.2	22.8	21.7	18.3	14.0	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	17.7	20.4	22.8	24.9	26.9	27.1	25.7	25.2	23.7	19.8	16.8
		MCDB	20.9	22.7	26.5	27.3	32.7	35.9	36.0	34.8	33.0	29.6	23.8	19.5
	2%	WB	12.5	15.4	18.8	21.1	23.7	25.4	25.7	25.0	24.3	22.4	17.9	13.9
		MCDB	18.1	20.7	24.2	26.8	30.8	33.8	34.5	34.2	31.5	28.2	22.2	17.2
	5%	WB	10.4	13.4	17.5	20.0	22.9	24.6	25.0	24.4	23.5	21.0	16.6	11.5
		MCDB	16.0	18.9	21.8	25.3	29.7	32.6	33.6	33.3	30.3	26.1	21.1	15.6
	10%	WB	8.6	11.3	15.7	18.9	21.9	24.0	24.3	23.8	22.8	19.8	15.0	9.2
		MCDB	14.6	16.6	20.5	23.7	28.3	31.7	32.6	32.6	29.4	24.6	19.4	14.3
Mean Daily Temperature Range	5% DB	MDBR	13.4	13.1	13.6	13.7	13.0	12.9	13.0	13.2	12.9	13.1	13.0	12.1
		MCDBR	18.8	18.5	17.4	17.3	16.0	15.3	15.3	15.5	15.2	16.1	16.5	16.8
		MCWBR	10.9	10.2	9.2	8.6	6.0	4.8	4.2	4.5	5.1	7.1	8.7	10.2
	5% WB	MCDBR	17.0	16.2	14.0	13.6	13.6	13.1	12.7	13.0	12.3	12.6	13.6	14.1
		MCWBR	10.7	10.3	8.5	8.0	6.3	5.1	4.4	4.2	4.4	6.7	8.5	9.8
Clear-Sky Solar Irradiance	taub	0.287	0.299	0.329	0.366	0.398	0.420	0.419	0.413	0.381	0.347	0.312	0.289	
	taud	2.530	2.504	2.439	2.357	2.340	2.311	2.358	2.319	2.382	2.465	2.497	2.522	
	Ebn at Noon	933	955	946	919	889	865	864	864	879	886	892	907	
	Edh at Noon	81	93	108	123	127	130	123	126	113	96	83	77	
All-Sky Solar Radiation	RadAvg	3.03	3.63	4.83	5.88	6.32	6.97	6.96	6.36	5.29	4.09	3.17	2.58	
	RadStd	0.30	0.50	0.30	0.46	0.66	0.59	0.41	0.37	0.39	0.56	0.40	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page